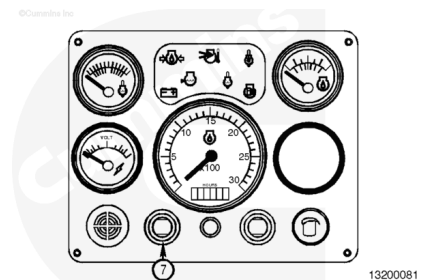


Trainings ▾

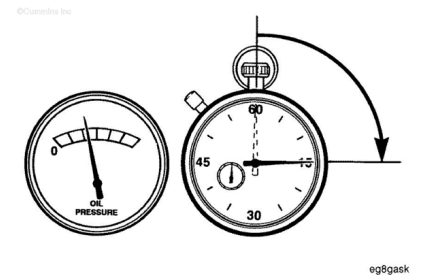
General Information

Note : There is a separate keyswitch wired to the primary panel. It will be installed in the helm at the boat manufacturer's or installer's desired location. The keyswitch **must** be in the ON position to crank-start the engine.

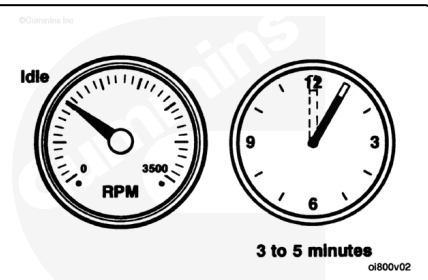
The engine start button (7) is used to engage the starter motor.



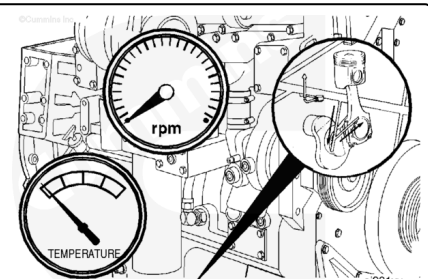
The engine **must** have adequate oil pressure within 15 seconds after starting. If the warning light indicating low oil pressure has **not** gone out or there is **no** oil pressure indicated on a gauge within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level in the oil pan.



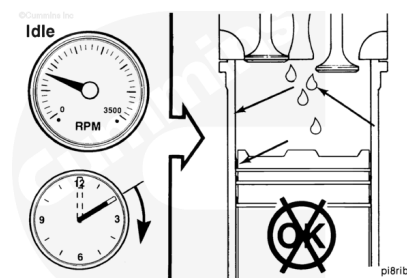
Idle the engine 3 to 5 minutes before operating with a load.



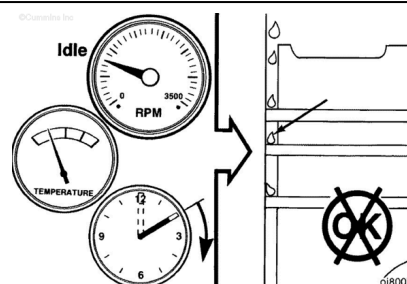
Increase the engine speed (rpm) slowly to provide adequate lubrication to the bearings and to allow the oil pressure to stabilize.



Do **not** keep the engine at low idle for long periods. Idling for periods longer than 10 minutes can damage an engine, causing combustion chamber temperatures to drop so low the fuel will **not** burn completely. This will cause carbon to build up around the injector spray holes and piston rings, and can cause the valves to stick.

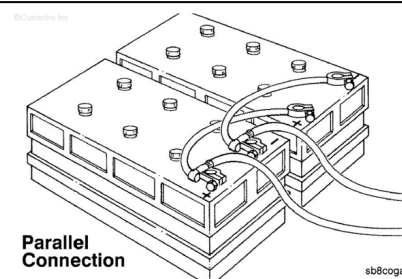


If the engine coolant temperature becomes too low, below 60° C [140° F], raw fuel will wash the lubrication oil off the cylinder walls and dilute the crankcase oil. Fuel dilution adversely affects lubricating oil properties and can shorten engine life. Utilize the fast idle to prevent these conditions.



⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ CAUTION ⚠

When using jumper cables to start the engine, make sure to connect the cables in parallel: Positive (+) to positive (+) and negative (-) to negative (-). When using an external electrical source to start the engine, turn the disconnect switch to the OFF position. Remove the key before attaching the jumper cables.

The accompanying illustration shows a typical parallel battery connection. This arrangement doubles the cranking amperage.

This illustration shows a typical series battery connection. This arrangement, positive (+) to negative (-), doubles the voltage.

